CSL7610 Assignment: A Classical Reasoning System for Logical Entailment

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1 Overview

The task is to decide whether a query is entailed by a knowledge base of facts and rules. The three public datasets ask for quite different kinds of reasoning: RuleTaker is about multi-step rule chaining with negation, FOLIO is full first-order logic, and CounterBench is about counterfactuals over causal graphs. One prover does not handle all three well, so the system has three engines and a small router in front of them:

$$\text{input (NL or logic)} \;\rightarrow\; \text{router} \;\rightarrow\; \begin{array}{l}\text{Horn rules} \rightarrow \text{stratified forward chaining}\\ \text{general FOL} \rightarrow \text{resolution refutation}\\ \text{causal statements} \rightarrow \text{structural causal model}\end{array} \;\rightarrow\; \text{True / False}$$

No language model is used anywhere. Converting text into the internal representation is done with a deterministic, grammar-based parser, and every prediction comes out of one of the three engines. The router decides by looking at the input itself and does not use dataset labels: FOL formulas go to the logic route, text that talks about causes and asks a "would …" question goes to the causal engine, and anything else goes to the rule parser. On the logic route, a theory that consists only of literals and rules of the form $\forall x\,(L_1 \wedge \cdots \wedge L_n \rightarrow L)$ is sent to forward chaining. Anything with $\vee$, $\oplus$, $\leftrightarrow$, $\exists$ or nested quantifiers goes to resolution.

Since the evaluation will give the logic directly, the system accepts several notations for formulas: Unicode ($\forall x\,(P(x) \rightarrow Q(x))$), ASCII (`forall x. (P(x) -> Q(x))`, `&`, `|`, `~` and Prolog clauses (`q(X) :- p(X), r(X).`).

2 Representation

Atoms and rules. An atom is a pair (predicate, argument tuple), for example `('young', ('bob',))` or `('eat', ('bear', 'squirrel'))`. A literal is an atom with a sign. A rule is a pair (body, head), where the body is a list of literals and the head is one literal. Variables start with `?`. Rules may contain negated body literals such as $\mathit{big}(X) \wedge \neg \mathit{red}(X) \rightarrow \mathit{kind}(X)$.

Text to rules. The RuleTaker parser works at the level of clauses. A clause is *subject* + *copula/modal/verb* + *predicate*. For example, "the bald eagle does not chase the cat" becomes $\neg \mathit{chase}(\mathit{bald_eagle}, \mathit{cat})$. Indefinites and pronouns (*someone*, *something*, *they*, *it*) become the variable X. Conjunctions with an elided subject ("is big and not red") are expanded by reusing the previous subject and copula. Generic sentences ("All big, red things are kind", "Kind people are smart", "Things that are rough are nice") become rules over X, and plural class nouns are singularised ("Penguins are birds" gives $\mathit{penguin}(X) \rightarrow \mathit{bird}(X)$). Verbs are lemmatised, so "chases" and "does not chase" map to the same predicate. On the full RuleTaker test split (151,911 questions) no sentence was left unparsed, including in the paraphrased NatLang theories.

First-order logic. Formulas are parsed by a recursive-descent parser (precedence $\neg \;>\; \wedge \;>\; \vee, \oplus \;>\rightarrow>\leftrightarrow$, quantifier scope extending to the right) into an AST. Conversion to clauses is the textbook sequence: eliminate $\leftrightarrow$, $\oplus$, $\rightarrow$; push negations inward (NNF); standardise variables apart; Skolemise; drop $\forall$; distribute $\vee$ over $\wedge$. Some FOLIO annotations have small syntax errors, such as an unbalanced bracket or a comma used for $\wedge$. These are repaired before parsing.

Causal models. CounterBench sentences such as "Glent causes Razz, Pex and Zurn, not Zurn causes Melf, and Melf and Razz together cause Zlim" are turned into Boolean structural equations: $\mathit{razz} := \mathit{glent}$, $\mathit{melf} := \neg \mathit{zurn}$, $\mathit{zlim} := \mathit{melf} \wedge \mathit{razz}$, and so on. "A or B causes C" gives a disjunction, and "A causes not B" a negated equation. `X~Bern(p)` gives a prior on a root variable, and "We observed X" gives evidence. The query ("Would Y occur if not X instead of X?", "if not A and not B", or the nested "Assume …, further suppose …") becomes a target variable, an intervention and the factual premise.

3 Reasoning Algorithms

3.1 Stratified forward chaining (RuleTaker and Horn logic)

Facts are stored in an index keyed by predicate. A rule fires when its positive body literals can be matched against known facts by unification. The matching is a backtracking search over body literals, which builds up the substitution θ. Rules are applied until a fixpoint is reached. Negated body literals use negation as failure: $\neg p(a)$ holds when $p(a)$ cannot be derived. To keep this sound, rules are first split into strata, so that a predicate defined through $\neg p$ is evaluated only after p is complete. Every derived fact stores the rule and substitution that produced it, and the proof traces in Section 5 are printed from this. A query is True if all its literals hold in the final model. Since a theory is shared by many queries, the closure is computed once per theory and cached.

3.2 Resolution refutation (FOLIO and general FOL)

To test $KB \models Q$, the prover adds the clauses of $\neg Q$ and searches for the empty clause. The search is a best-first given-clause loop. The set of support is initialised with the negated-goal clauses, and the shortest clause is always picked next (unit preference). Each picked clause is resolved against the already-processed clauses through a (sign, predicate) index. The loop uses binary resolution with most general unifiers (with occurs check), factoring, tautology deletion, duplicate removal via canonical variable renaming, and forward subsumption by unit clauses. First-order resolution need not terminate, so the search has limits: 6 literals per clause, Skolem term depth 3, 4,000 clauses or 1 second. If no refutation is found within these limits, the answer is False. A FOLIO conclusion is labelled True only when it is provable; both "False" and "Uncertain" count as not entailed.

3.3 Counterfactual inference in a structural causal model (CounterBench)

Counterfactuals are answered with the three steps of Pearl's procedure. *Abduction:* all assignments of the root variables are enumerated, and those inconsistent with the evidence and the factual premise ("instead of X" means X was true) are discarded. Each remaining world is weighted by the Bernoulli priors, with 0.5 for unspecified roots. *Action:* the intervened variables are fixed with $do(\cdot)$. *Prediction:* the equations are evaluated in topological order with memoisation. The answer is True when the posterior probability of the target is above 0.5. In the deterministic cases this probability is either 0 or 1.

4 Experiments on the Public Data

Setup. RuleTaker: the complete `tasksource/ruletaker` test split (151,911 questions, depth-0 to depth-5 and both NatLang configurations). During development I worked on a random sample of 5,000 of these (seed 42) and ran the full split once at the end. FOLIO: the official validation set (204 examples, gold FOL annotations). CounterBench: the two balanced data files (1,200 questions in five types). All times are wall-clock times on a single CPU core of the institute login node and include parsing.

Table 1: Main results. For CounterBench the held-out number is the one to compare (see Sec. 4.3).

Dataset	Queries	Accuracy (%)	Time (ms / query)
RuleTaker (text)	151,911	99.97	0.075
RuleTaker (same theories given as logic)	151,911	99.97	0.72
FOLIO (FOL annotations)	204	98.04	19.3
CounterBench (held-out half)	600	94.17	0.21
All three datasets	153,315	99.93	total 15.5 s

4.1 Closed vs. open world for RuleTaker

The labels of RuleTaker depend on how negation is read, so I ran the same engine under both semantics (Table 2). Under the open-world reading, $\neg p$ needs an explicitly derived negation and accuracy drops to about 75% at every depth (74.57% overall). Under the closed-world reading with stratified negation as failure, 41 of

the 151,911 questions are wrong (99.97%), and almost all of them are at depth 3 or deeper. The closed-world semantics is therefore used for Horn-style theories. Feeding the same theories as logic formulas (for example $\forall x\,(\mathit{Young}(x) \wedge \neg \mathit{Big}(x) \rightarrow \mathit{Smart}(x))$) through the logic route gave exactly the same 99.97%, with the same per-depth numbers, which checks that the logic route behaves the same as the text route.

Table 2: RuleTaker accuracy (%) by configuration, full test split.

	depth-0	depth-1	depth-2	depth-3	depth-3ext	depth-5	NatLang	3ext-NatLang
n	17347	17788	17741	17982	23222	19021	7793	31017
Open world	77.94	74.15	73.61	73.46	73.71	74.97	75.67	74.23
Closed world (NAF)	100.0	100.0	99.99	99.96	99.97	99.92	100.0	99.97

4.2 FOLIO

Resolution proves 68 of the 72 True conclusions and produces no false positives: all 132 False or Uncertain conclusions are correctly not proved. I re-ran the four misses with a 20 times larger search budget. Three of them still saturate without a proof, which means the gold FOL does not entail them. In one case this is just operator precedence: "$\mathit{MountainRange}(p) \wedge \mathit{In}(p, \mathit{newmexico}) \vee \mathit{In}(p, \mathit{texas})$" makes $\mathit{MountainRange}(p)$ part of only the first disjunct, so the conclusion that Juan de Onate visited a mountain range in New Mexico cannot be derived. The fourth (the Sir Digby problem, which contains an $\leftrightarrow$ over two-place predicates) is still open when the larger budget runs out. Fourteen FOLIO problems are pure Horn theories and went to forward chaining; all fourteen are correct.

4.3 CounterBench: reading conventions and a held-out check

CounterBench took the most iterations, and most of the gains came from reading the wrong predictions one by one. The first version scored 72.9%. Looking at the errors showed three parser bugs. Nested questions ("Assume not X, and based on this assumption, further suppose not Y") were never reaching the causal engine, because the pattern was matching the "Imagine a self-contained world" preamble. "A and B causes C" was being split as if it were two causal statements. Some observations were run together with the next sentence because of a missing full stop. Fixing these brought the score to 90.5%.

The remaining errors were about how the questions are worded. Take "P causes not Q". When an observation or a question later says "not Q", it can be read as $Q = 0$, or as naming the negated node itself. I first applied the second reading everywhere because it fixed the five examples I was looking at. On all 1,200 questions it lost (75.8% vs. 79.6% at that stage), so a global switch was the wrong choice. I therefore made the reading a parameter with five binary or ternary options (negation reading for observations and for interventions, clamping observed values, the answer under inconsistent evidence, and how multiple interventions combine; 32 combinations). The best combination is chosen separately for each question type, and the type is detected from the text. To avoid fitting the evaluation data, the questions are split into two halves: the convention is chosen on one half and accuracy is reported on the other.

Table 3: CounterBench accuracy (%) on the held-out half, by question type.

	basic	conditional	joint	nested	backdoor	all
n	122	129	136	119	94	600
Default reading	100.0	65.89	98.53	95.80	93.62	90.50
Tuned per type	100.0	82.95	98.53	95.80	93.62	**94.17**

The split also caught one case of overfitting. For *joint* questions, one convention beat the default by a single example on the selection half and then lost 4.4 points on the held-out half. After this, a type only switches when the gain on the selection half is at least 3 examples or 2%. With this rule only *conditional* and *backdoor* switch, and no type is worse than the default on held-out data. Most of the remaining errors cannot be explained by the stated mechanisms. One example: "Blorn and Plim together cause not Quaz, Quaz causes Skul, Skul causes Jext; we observed not Quaz" is labelled *yes* for "Would Jext occur if not Fizo?", but Jext depends on Fizo through no path. I did not try to fit such cases.

4.4 Efficiency

Forward chaining and the causal engine take well under a millisecond per query; the whole RuleTaker test split (151,911 questions) runs in 11.3 seconds. Caching the closure per theory matters here, because many questions share a theory, and it is also why the per-query time on the full split (0.075 ms) is lower than on the 5,000 sample (0.41 ms). Parsing the logic form is slower than the text form (0.72 ms per query), mostly because each formula is tokenised and parsed separately. Resolution is the expensive part at about 19 ms per query, and most of that time is spent on non-entailed conclusions, which run until the search limits are reached. The clause and time limits are what keep the total time low: with them, the whole FOLIO validation set takes about 4 seconds.

5 Reasoning Traces

Forward chaining (RuleTaker, depth-5, query "Harry is blue", True). Sub-proofs that have already been shown are not repeated.

```
blue(harry)    [rule: green(X) ∧ white(X) → blue(X)  θ={X/harry}]
    green(harry)   [rule: nice(X) → green(X)  θ={X/harry}]
        nice(harry)   [fact]
    white(harry)   [rule: red(X) → white(X)  θ={X/harry}]
        red(harry)   [rule: rough(harry) ∧ round(harry) → red(harry)  θ={}]
            rough(harry)   [rule: nice(X) → rough(X)  θ={X/harry}]
                nice(harry)   [fact]
            round(harry)   [rule: rough(X) ∧ green(X) → round(X)  θ={X/harry}]
                rough(harry)   [proved above]
                green(harry)   [proved above]
```

Resolution refutation (FOLIO, True; b = bonnie). The conclusion is

$$(\mathit{AcademicCareer}(b) \oplus \mathit{Chaperone}(b)) \rightarrow (\mathit{AcademicCareer}(b) \oplus \mathit{Inactive}(b)).$$

The negated goal is split into clauses and resolved down to the empty clause.

```
¬academiccareer(b) ∨ ¬chaperone(b)      [negated goal]
¬inactive(V0) ∨ chaperone(V0)           [premise]
¬academiccareer(b) ∨ ¬inactive(b)       ⇐ resolve of the two lines above
¬inactive(b) ∨ academiccareer(b)        [negated goal]
¬inactive(b)                            ⇐ resolve with ¬academiccareer(b) ∨ ¬inactive(b)
inactive(V0) ∨ talentshows(V0)          [premise]
talentshows(b)                          ⇐ resolve with ¬inactive(b)
¬talentshows(V0) ∨ engaged(V0)          [premise]
engaged(b)                              ⇐ resolve with talentshows(b)
¬academiccareer(b) ∨ inactive(b)        [negated goal]
¬academiccareer(b)                      ⇐ resolve with ¬academiccareer(b) ∨ ¬inactive(b)
academiccareer(b) ∨ chaperone(b)        [negated goal]
¬chaperone(V0) ∨ ¬students(V0)          [premise]
¬students(b) ∨ academiccareer(b)        ⇐ resolve of the two lines above
¬students(b)                            ⇐ resolve with ¬academiccareer(b)
¬engaged(b) ∨ students(b)               [premise]
¬engaged(b)                             ⇐ resolve with ¬students(b)
□  (empty clause)                       ⇐ resolve engaged(b) with ¬engaged(b)
```

Counterfactual (CounterBench, nested: "Assume not Glent, and based on this assumption, further suppose not Melf. Would Zlim occur?", gold *no*).

```
razz := glent    pex := razz    zurn := pex    melf := zurn    zlim := melf
do(glent = 0), do(melf = 0)   ->   zlim = melf = 0   ->   prediction: False
```

6 Limitations

- The text parser is pattern based. It covers the RuleTaker templates and their NatLang paraphrases, but not arbitrary English. For FOLIO, the system depends on FOL being supplied, which the evaluation does.

- Resolution is run with limits, so a true entailment with a very long proof can be reported as False. One of the four FOLIO misses may be of this kind.
- The CounterBench reading conventions were selected on public data. They are chosen per detected question type rather than per example, and validated on a held-out half, but questions worded in a new way could still need a different reading.
- The output is binary, so FOLIO's *False* and *Uncertain* are both mapped to not entailed.